Kunal Agarwal

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PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienced in designing fault-tolerant microservices, optimizing database throughput, and shipping production code with rigorous test coverage.

EDUCATION

B.Tech in Electronics and Communication Engineering — Jadavpur University *2020 - 2024*

Cumulative Grade Point Average: 9.10/10.0

WORK EXPERIENCE

Platform Engineer — Swiggy (Bundl Technologies), Bengaluru, India *Jun 2023 - Present*

- Implemented automated circuit-breaker pattern with Resilience4j across 8 downstream microservices, preventing cascade outages during third-party gateway downtime.
- Architected high-throughput payment webhook processor using Go and Redis Streams, handling 14,000 requests/sec with p99 latency under 28ms.
- Developed responsive internal operations dashboard in React and TypeScript with TanStack Query, accelerating merchant dispute resolution by 40%.

TECHNICAL PROJECTS

Peer-to-Peer File Transfer Protocol with NAT Traversal | Technologies: Rust, STUN/TURN, UDP, Tokio, Cryptography

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.

High-Concurrency Real-Time Collaboration Canvas | Technologies: TypeScript, WebSocket, CRDT, React, Node.js

- Built multi-user collaborative workspace using Conflict-free Replicated Data Types (CRDTs) to ensure convergence across disconnected edits.
- Engineered WebSocket server in Node.js with delta-compression protocol, reducing network bandwidth consumption by 64% during intense cursor movements.
- Authored comprehensive property-based test suite with fast-check verifying eventual consistency across simulated random network partitions.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines